

Optimizing cerebral oxygenation in cardiac surgery: A randomized controlled trial examining neurocognitive and perioperative outcomes



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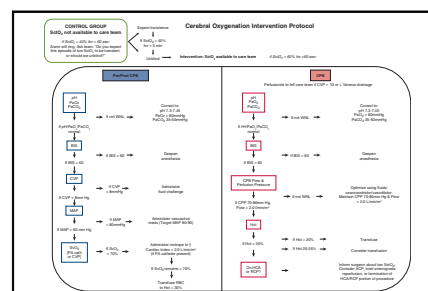
ABSTRACT

Objective: The study objective was to determine whether targeted therapy to optimize cerebral oxygenation is associated with improved neurocognitive and perioperative outcomes.

Methods: In a prospective trial, intraoperative cerebral oximetry monitoring using bilateral forehead probes was performed in cardiac surgical patients who were randomly assigned to an intervention group in which episodes of cerebral oxygen desaturation (<60% for >60 consecutive seconds at either probe) triggered an intervention protocol or a control group in which the cerebral oximetry data were hidden from the clinical team, and no intervention protocol was applied. Cognitive testing was performed preoperatively and at postoperative months 3 and 6; domains studied were response speed, processing speed, attention, and memory. Perioperative outcomes studied were death, hospital length of stay, intensive care unit length of stay, postoperative day of extubation, time on mechanical ventilation, intensive care unit delirium, Sequential Organ Failure Assessment on intensive care unit admission, and intensive care unit blood transfusion.

Results: Group mean memory change scores were significantly better in the intervention group at 6 months (0.60 [standard error, 0.30] vs -0.17 [standard error, 0.33], adjusted $P = .008$). However, presence, duration, and severity of cerebral desaturation were not associated with cognitive change scores. Perioperative outcomes did not differ between the intervention and control groups.

Conclusions: Targeted therapy to optimize cerebral oxygenation was associated with better memory outcome in a group of cardiac surgical patients. Some aspects of the protocol other than desaturation duration and severity contributed to the observed neuroprotective effect. (J Thorac Cardiovasc Surg 2020;159:943-53)



Intervention protocol for intraoperative cerebral oxygenation desaturation episodes.

Central Message

In cardiac surgical patients, a NIRS-guided treatment algorithm was associated with better cognitive outcome, although presence, duration, and severity of desaturation were not predictors.

Perspective

NIRS monitoring and a treatment algorithm had a neuroprotective effect on cognitive outcome in a group of cardiac surgical patients. Presence, duration, and severity of cerebral desaturation per se were not associated with neurocognitive outcome, indicating that aspects of the treatment algorithm for cerebral oxygenation optimization were contributory.

See Commentaries on pages 954 and 956.

Cardiac surgery has been associated with postoperative cognitive decline (POCD).¹ Multiple factors have been found to contribute, including cardiopulmonary bypass

(CPB) parameters, temperature management, acid-base management, hemoglobin concentration and transfusion, embolization of gaseous and particulate matter during CPB, and cerebral hypoperfusion, all of which might result in a mismatch between cerebral oxygen supply and demand. Cerebral oximetry based on near-infrared spectroscopy

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Abbreviations and Acronyms

CI	= confidence interval
CPB	= cardiopulmonary bypass
NIRS	= near-infrared spectroscopy
POCD	= postoperative cognitive decline
rSO ₂	= regional cerebral oxygen saturation
WGEE	= weighted generalized estimating equation

(NIRS) technology measures regional saturation of oxygen (rSO₂) at the microvascular level.² It is continuous, noninvasive, available at the point of care, and a sensitive index of cerebral hypoxia or cerebral ischemia, the main causes of brain injury in cardiac surgical patients.^{3,4}

NIRS oximetry monitoring holds promise for guiding interventions to improve intraoperative brain oxygenation and clinical outcomes, particularly POCD. A meta-analysis of 4 randomized controlled trials performed in cardiac surgical patients found that intraoperative management guided by cerebral oximetry was associated with reduced incidence of POCD (risk ratio, 0.55; 95% confidence interval [CI], 0.36-0.86; $P = .0009$; $I^2 = 85\%$) at hospital discharge or 1 week postoperatively.⁵

This prospective randomized controlled trial in patients undergoing cardiac surgery with CPB investigated whether a treatment algorithm targeted to optimize intraoperative cerebral oxygenation intervention (“targeted therapy”) was associated with better neurocognitive and perioperative outcomes compared with standard care. Cognitive outcomes studied were response speed, processing speed, attention, and memory. Perioperative outcomes studied were death, hospital length of stay, intensive care unit (ICU) length of stay, postoperative day of extubation, time on mechanical ventilation, ICU delirium, Sequential Organ Failure Assessment on ICU admission, and ICU blood transfusion.

MATERIALS AND METHODS

This study was conducted under an Icahn School of Medicine at Mount Sinai Institutional Review Board–approved protocol with informed consent.

Patients

Adult patients (aged >18 years) undergoing elective cardiac surgery requiring CPB were studied. The exclusion criteria were (1) severe preoperative cognitive impairment by history or preoperative cognitive testing; (2) sensory or motor impairment precluding reliable operation of a computer; (3) lack of access to computer-based cognitive evaluation; (4) non-English-speaking patients; (5) renal failure requiring dialysis; (6) respiratory failure requiring home oxygen use; (7) Child’s B or C hepatic failure; (8) procedures with deep hypothermic circulatory arrest; and (9) baseline oximetry values less than 60%. Eligible patients who consented to participate and met the enrollment criterion of completing preoperative cognitive testing were assigned to an intervention or control group differing with respect to rSO₂ treatment algorithms (described next) by simple randomization using an online true random number generator (www.random.org).

Anesthetic Management

After placement of a preinduction arterial line, general anesthesia was induced with midazolam, fentanyl, etomidate, or propofol, and a muscle relaxant. A balanced anesthetic technique including isoflurane, fentanyl, and a muscle relaxant was used for maintenance of anesthesia. Patients with pulmonary hypertension or ventricular dysfunction were monitored with a pulmonary artery catheter. Central venous pressures, pulmonary artery pressures, and transesophageal echocardiography were used to guide fluid management. Standard blood conservation protocols were followed.

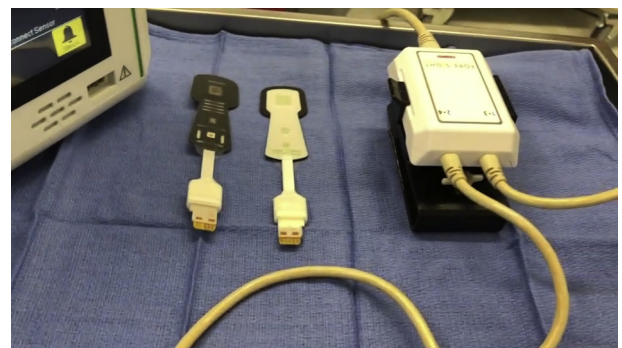
Cardiopulmonary Bypass Technique

Cannulation varied according to the needs of the surgical procedure. Temperature was allowed to drift to a nadir of 31° to 32° during CPB. Perfusion flow was maintained at or close to 2.2 L/min/m². Perfusion pressure was adjusted using a phenylephrine infusion to maintain a mean arterial pressure of 60 to 80 mm Hg. End-tidal isoflurane concentration or intravenous drugs (fentanyl and midazolam) were titrated to maintain a bispectral index value less than 60. By using hemofiltration and autologous red blood cell transfusions, the hematocrit was maintained at greater than 20%.

Cerebral Oximetry Monitoring and Intervention Protocol

Cerebral oxygenation was monitored using the FORE-SIGHT cerebral oximeter (CAS Medical Systems, Branford, Conn), which measures absolute regional oxygen saturation using continuous-wave spatially resolved NIRS by probes placed on the forehead above the frontal lobes bilaterally (Video 1).^{6,7} Oximetry monitoring was initiated in the operating room before anesthetic induction and recorded continuously every 2 seconds throughout surgery. Baseline data were obtained for the first 2 minutes after placement of the cerebral monitors before sedation while patients were breathing room air or receiving low flow rates of nasal cannula oxygen. Because the Foresight is an absolute cerebral oximeter, baseline data were obtained to ensure that the starting value for each patient was above the threshold for intervention (ie, ≥60%). Patients with baseline values less than 60% were excluded from the study.

In the intervention group, the oximetry data were displayed openly to the perioperative team. The criterion for intervention was defined as rSO₂ less than 60% for more than 60 consecutive seconds at either probe. An rSO₂ less than 60% triggered an alarm, the anesthesiologist timed the event, and after 60 seconds initiated an intervention protocol similar to that of Denaut and colleagues⁸ until the rSO₂ was restored to at least 60% at both probes (Figure 1). For repeated cerebral desaturation episodes, the treatment algorithm was reinitiated, as appropriate to phase of care.



VIDEO 1. Cerebral oximetry demonstration and study explanation on a standard model patient. Video available at: [https://www.jtcvs.org/article/S0022-5223\(19\)30710-X/fulltext](https://www.jtcvs.org/article/S0022-5223(19)30710-X/fulltext).

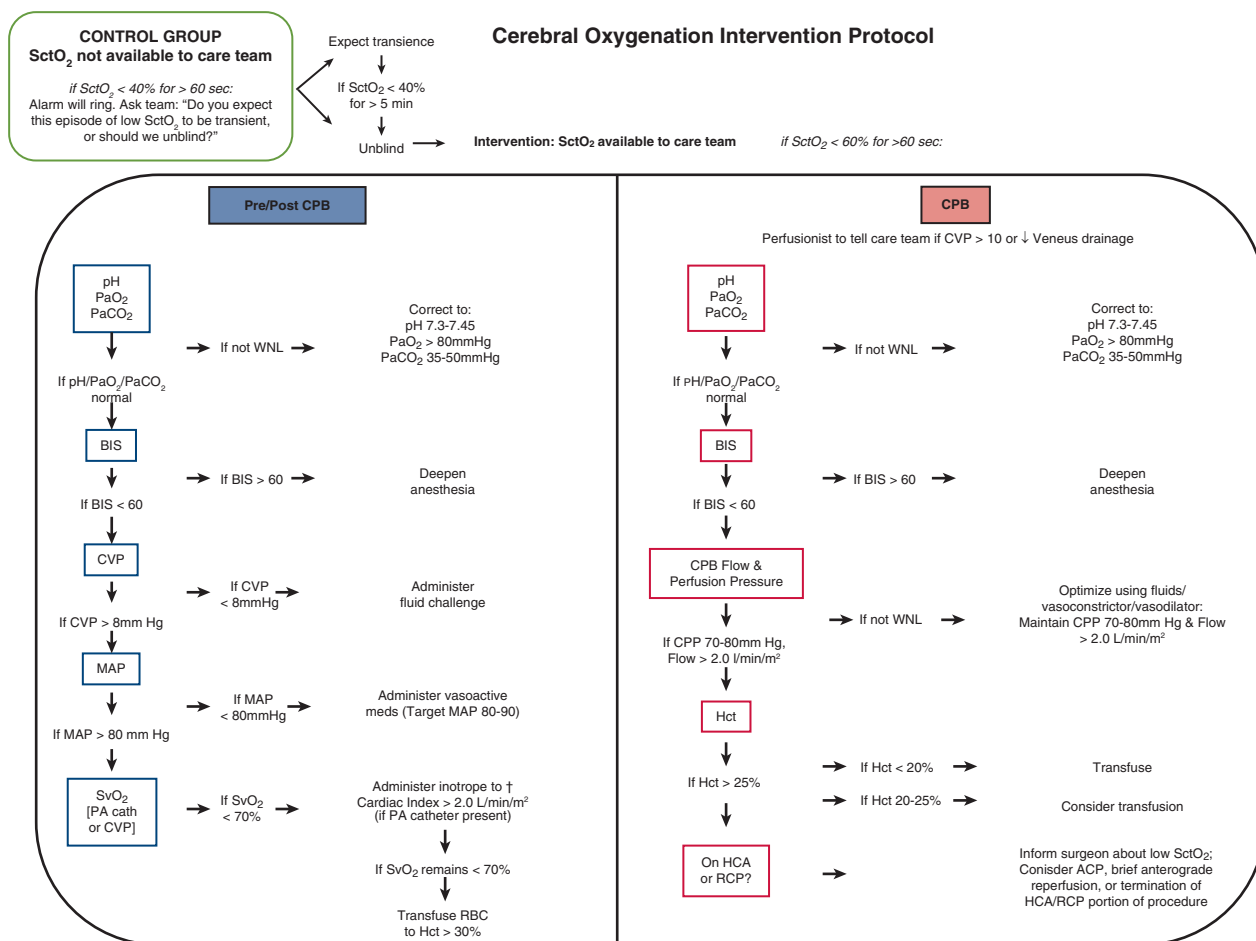


FIGURE 1. Intervention protocol. The protocol for cerebral oxygenation-driven interventions is illustrated. As described in the upper left corner, in the control group, rSO₂ data were hidden from the clinical team unless a dangerously low value (<40%) was present. The remainder of the diagram demonstrates the algorithm for treating patients with episodes of cerebral desaturation (rSO₂ <60%) occurring during the pre- and post-CPB periods (left) and during CPB (right). Interventions specified in the algorithm optimized oxygenation, ventilation, depth of anesthesia, cerebral perfusion pressure, and hemoglobin-oxygen carrying capacity. CPB, Cardiopulmonary bypass; WNL, within normal limits; pH, potential hydrogen; BIS, bispectral index; CVP, central venous pressure; MAP, mean arterial pressure; PA, pulmonary artery; RBC, red blood cells; CPP, cerebral perfusion pressure; Hct, hematocrit; HCA, hypothermic circulatory arrest; ACP, antegrade cerebral perfusion; RCP, retrograde cerebral perfusion.

In the control group, the oximetry data were concealed from the care team by a white screen covering the monitor display. The protocol allowed for revealing the oximetry data in the event of a critically low value, defined as rSO₂ less than 40% for more than 60 seconds at either probe, that was signaled by an alarm. If the episode was expected to be transient, revealing the oximetry data and intervention would be postponed until rSO₂ remained less than 40% for more than 5 minutes. If the episode was not expected to be transient, the patient would cross over into the intervention group; rSO₂ data would be revealed to the clinical team for the remainder of the operative period, and the intervention protocol would be instituted until rSO₂ increased to more than 60%.

Data Collection

Patients underwent cardiac surgery from November 30, 2011, to October 20, 2014. Cognitive testing data collection occurred from November 23, 2011, to December 14, 2014.

Eligible patients were recruited at their surgical preadmission testing appointment. Patients who consented to participate and completed

preoperative internet-based cognitive testing were enrolled in the study. We proposed to enroll patients until 100 completed all 3 phases of cognitive testing; however, the study was discontinued prematurely after 75 patients completed all 3 phases of cognitive testing (with oximetry and perioperative outcome data collected on 125 patients), because the test vendor went out of business.

Preoperative patient data. Demographic data collected were age, gender, and educational level. Preoperative risk factors for adverse outcomes were body mass index (BMI), European System for Cardiac Operative Risk Evaluation, and ejection fraction.

Operative data. Surgical procedure was classified into 5 types: coronary artery bypass grafting, single valve, multiple valves, valve and other cardiac procedure, and other cardiac procedure (ie, aortic root replacement, ventriculomyotomy). Surgery and CPB durations were recorded.

Cerebral oximetry data. The 2 streams of oximetry data were converted into a single stream of data by taking the lowest value at each 2-second interval (unless there was a loss of data from 1 probe). In the situation where there was loss of data from both probes, data were imputed on

the basis of the assumption that the rSO_2 changed linearly during the interval of missing data (ie, by calculating the slope between the rSO_2 data points that occurred before and after the interval of missing data).

Cerebral oximetry data parameters were baseline rSO_2 , incidence of desaturation, duration of desaturation, and severity of desaturation. For baseline oximetry, the average rSO_2 value over the 2-minute interval before anesthetic induction was computed for each probe; the lower of the 2 values was defined as the patient's baseline. Incidence of desaturation was defined as the number of patients who experienced at least 1 episode of desaturation ($<60\%$ for >60 consecutive seconds). Desaturation duration was quantified as the cumulative amount of time under the 60% threshold (minutes) that occurred within desaturation episodes. Desaturation severity was quantified as the "area under the threshold" (integration of desaturation percentage below 60% over time, %-minutes) that occurred within desaturation episodes. Because the desaturation data were skewed (even when patients with no desaturation episodes were excluded), we also performed log-transformation for the subset of patients with desaturation.

Neurocognitive outcomes. Internet-based cognitive testing was performed using the Cognitive Stability Index HeadMinder battery

(HeadMinder, Inc, New York, NY) at 3 time points: (T1) preoperatively (>24 hours before surgery), (T2) 3 months postoperatively, and (T3) 6 months postoperatively. The Cognitive Stability Index battery consists of 10 subtests that resolve into 4 factors: response speed, processing speed, attention, and memory.⁹ Test scores that were flagged as invalid (ie, extreme scores that preclude meaningful interpretation and are likely confounded by extraneous factors) were removed from the data. Cognitive outcomes were defined as the change in 4 cognitive factor scores from preoperative to postoperative time points. For memory and attention, higher test scores reflect better test performance. For response speed and processing speed factors, lower test scores reflect better test performance; thus, the sign was adjusted so that negative scores always reflected a preoperative to postoperative decline in performance, and positive scores always reflected an improvement in performance. As timed tests, the distributions of response speed and processing speed scores were highly skewed; thus, those scores were log-transformed.

Perioperative outcomes. Perioperative outcomes studied were death, hospital length of stay (days), intensive care unit (ICU) length of stay (days), extubation (day), time on mechanical ventilation (hours),

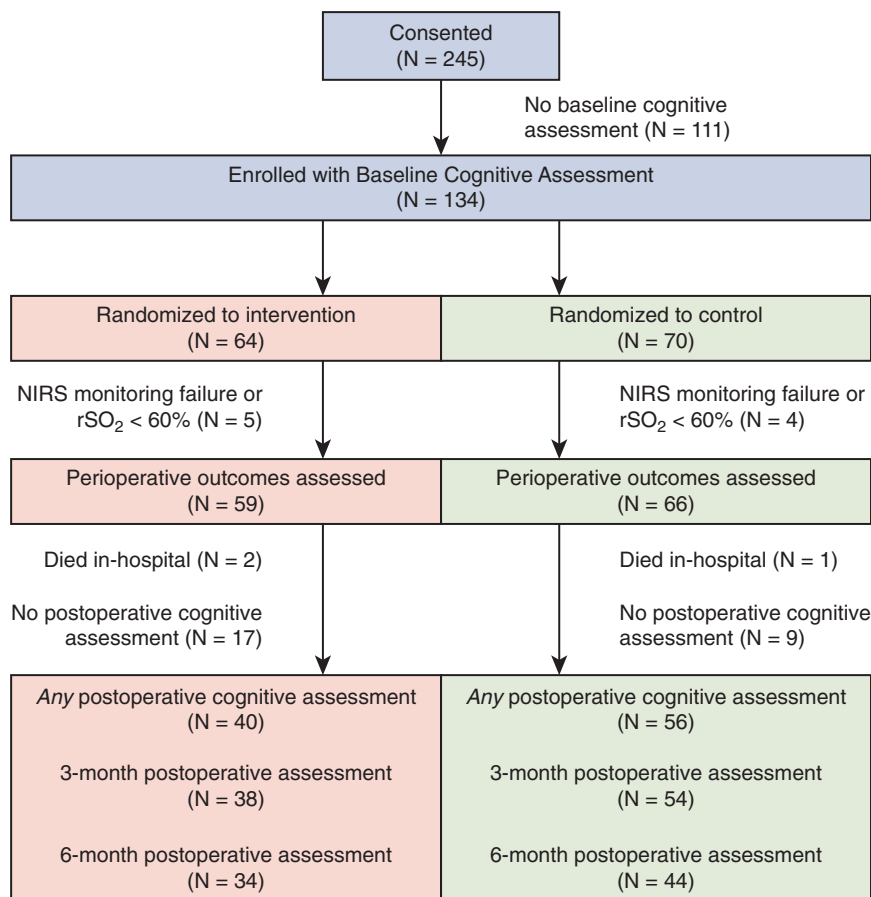


FIGURE 2. CONSORT flow diagram. A total of 245 patients consented to participate in the study. Of these patients, 134 completed baseline cognitive testing and were enrolled in the study and randomized to the intervention group ($N = 64$) or control group ($N = 70$). No patients were dropped from the study because of NIRS monitoring failures (including baseline values $<60\%$), leaving 125 patients (59 in the intervention group and 66 in the control group) available for studying perioperative outcomes. Of the 59 patients assigned to the intervention group, 2 died in the hospital, 38 completed the 3-month postoperative cognitive assessment, and 34 completed the 6-month postoperative cognitive assessment. Of the 66 patients assigned to the control group, 1 died in the hospital, 54 completed the 3-month postoperative cognitive assessment, and 44 completed the 6-month postoperative cognitive assessment. *NIRS*, Near-infrared spectroscopy; rSO_2 , regional cerebral oxygen saturation.

ICU delirium, Sequential Organ Failure Assessment on ICU admission, and ICU blood transfusion.

Data Analysis

Because the trial ended early and the 2 groups were not balanced (described next), we evaluated the similarity between the intervention and control groups with respect to demographics, risk factors, and operative characteristics. Because there were no significant differences between the groups, we compared their cerebral oximetry and perioperative outcomes using simple tests without adjustment for covariates. Chi-square or Fisher exact tests were used for categorical variables, Student *t* tests were used for normally distributed continuous variables, and Kolmogorov–Smirnov tests were used for non-normally distributed continuous or rank variables.

The primary objective of the analysis was to compare the differences between the 2 groups with respect to oximetry data, cognitive outcome data, and perioperative outcome data. For the cognitive data, change scores from the preoperative and postoperative time points (ie, T1 to T2, and T1 to T3) were compared between the groups using 2 approaches: (1) the analysis of covariance method that controlled for the baseline score; and (2) the weighted generalized estimating equation (WGEE) method that modeled the repeated measures of cognitive data accounting for subjects who were lost to follow-up. *P* values were adjusted for multiple comparisons for the 4 cognitive outcomes at each postoperative test using the false discovery rate method.¹⁰ A secondary analysis of the cognitive data was performed to evaluate the relationship between the cognitive change scores and oximetry data using the WGEE method. Details of the statistical models and results of the secondary analysis are described in the [Appendix E1](#). All statistical analyses were carried out using SAS (SAS Institute, Inc, Cary NC).

RESULTS

Patient Participation

[Figure 2](#) presents a CONSORT flow diagram for the study. A total of 245 patients enrolled in the study. A total of 134 patients completed preoperative testing; 64 (48%) were randomly assigned to the intervention group, and 70 (52%) were randomly assigned to the control group. Nine patients were removed from the study because of NIRS monitoring failures and low (<60%) baseline rSO₂ values. Of the 125 patients who successfully underwent intraoperative NIRS monitoring, 47% (59) were in the intervention group and 53% (66) were in the control group preoperatively; perioperative outcome data are reported for this sample. Three patients died before hospital discharge.

Ninety-six of the surviving 122 patients (79%) completed at least 1 postoperative cognitive test, with 92 (75%) and 78 (64%) completing postoperative testing at 3 and 6 months, respectively. Discontinuation of the test by the vendor contributed to the loss to follow-up. Of the 96 patients with postoperative cognitive test data, 40 (42%) were in the intervention group and 56 (58%) were in the control group, with a significantly greater dropout rate in the intervention group (chi-square test, *P* = .024). There were no statistically significant differences in terms of demographics, risk factors, operative

TABLE 1. Demographics, risk factors, and operative characteristics of the control and intervention groups

Variable	Intervention group (N = 59)		Control group (N = 66)	
	Mean	(SD) [Q1, Q3] %	Mean	(SD) [Q1, Q3] %
	Median		Median	
	N		N	
Demographics				
Age, y	57	(11)	58	(12)
Male gender	40	68%	46	70%
Years of education				
≤12 y	2	3.4%	1	1.5%
13-15 y	13	22.0%	20	30.3%
≥16 y	44	74.6%	45	68.2%
Risk factors				
BMI	27.1	(5.3)	26.6	(5.2)
euroSCORE	2.8	(3.11)	2.3	(2.34)
Ejection fraction	60	[55, 65]	60	[55, 65]
Operative characteristics				
Surgery duration (min)	296	[263, 345]	301	[258, 367]
CPB duration (min)	130	[108, 160]	132	[108, 174]
Surgery classification				
CABG	3	5.1%	4	6.1%
Single valve	9	15.3%	17	25.8%
Multiple valve	27	45.8%	31	47.0%
Valve and other procedure	18	30.5%	14	21.2%
Other procedure	2	3.4%	0	0%

SD, Standard deviation; BMI, body mass index; euroSCORE, European System for Cardiac Operative Risk Evaluation; CPB, cardiopulmonary bypass; CABG, coronary artery bypass grafting.

TABLE 2. Oximetry data

Variable	Intervention			Control			Unadjusted <i>P</i> value	FDR adjusted <i>P</i> value
	Mean	(SD) [Q1, Q3]	Mean	(SD) [Q1, Q3]				
	Median		Median					
	N		N					
Oximetry data for all patients (N = 125)								
Baseline rSO ₂	59	72	[69, 74]	66	73	[70, 76]	.331 ^a	
rSO ₂ desaturation incidence	59	39	66.1%	66	35	53.0%	.138 ^a	
Desaturation duration (min)	59	3	[0, 23]	66	1	[0, 26]	.423 ^a	.423 ^a
Desaturation severity (%-min)	59	8	[2, 51]	66	2	[0, 66]	.039^a	.078 ^a
Oximetry data of the subgroup of patients with desaturation (N = 74)								
Desaturation duration (min)	39	15	[3, 35]	35	24	[8, 67]	.363 ^a	.363 ^a
Desaturation severity (%-min)	39	29	[8, 112]	35	62	[27, 225]	.122 ^a	.257 ^a
Desaturation duration, log-transformed (min)	39	2.50	(1.44)	35	2.94	(1.44)	.193 ^b	.257 ^b
Desaturation severity, log-transformed (%-min)	39	3.52	(1.63)	35	4.04	(1.77)	.193 ^b	.257 ^b
Oximetry data of the subgroup of patients with desaturation and cognitive outcome data (N = 56)								
Desaturation duration (min)	26	12	[3, 32]	30	28	[9, 69]	.109 ^a	.109 ^a
Desaturation severity (%-min)	26	26	[8, 69]	30	73	[29, 300]	.013^a	.040^a
Desaturation duration, log-transformed (min)	26	2.43	(1.22)	30	3.06	(1.48)	.095 ^b	.109 ^b
Desaturation severity, log-transformed (%-min)	26	3.31	(1.44)	30	4.29	(1.61)	.020^b	.040^b

Bold indicates $P < .05$. *P* values are based on a: Kolmogorov–Smirnov test, b: *t* test. The *P* value adjustment for false discovery rate is within each panel. Episodes of cerebral oxygen desaturation were defined as less than 60% for more than 60 consecutive seconds at either oximetry probe. Desaturation duration was quantified as the cumulative amount of time under the 60% threshold (minutes) that occurred within desaturation episodes. Desaturation severity was quantified as the “area under the threshold” (integration of desaturation percentage <60% over time, %-minutes) that occurred within desaturation episodes. Because the desaturation data were skewed, we also performed log-transformation for the subset of patients with desaturation. The top panel shows data for the entire sample of 125 patients (ie, patients with and without desaturation). Baseline oximetry data and incidence of cerebral desaturation (ie, the number of patients who experienced at least 1 episode of desaturation) did not differ between the intervention and control groups. Duration of desaturation also did not differ between the groups. Median severity of desaturation was slightly greater in the intervention group, but this difference did not remain statistically significant after adjusting for multiple comparisons. The middle panel shows data for patients who had at least 1 desaturation episode. As expected, desaturation duration and severity were lower in the intervention group, but the differences were not statistically significant. The bottom panel shows data for patients with desaturation who had postoperative cognitive testing. Desaturation duration and severity were lower in the intervention group. Desaturation duration did not differ significantly, but desaturation severity did. With log-transformation, the normality assumption was met, and *t* test results agreed with the nonparametric test results. *FDR*, False discovery rate; *SD*, standard deviation.

characteristics, or oximetry data between the patients who did and patients who did not complete postoperative testing.

Preoperative Patient Data and Operative Data

Demographic, risk factor, and operative characteristics of the patient sample are presented in Table 1; there were no differences between the intervention and control groups.

Cerebral Oximetry Data

No patients in the control group fell below the critically low value of rSO₂ less than 40% for more than 60 seconds; thus, no cases crossed over into the intervention group. The median (interquartile range) percentage of time per case that required rSO₂ data imputation was 0.34% (0.13%–0.78%) of the total recording time from baseline to the end of the surgical procedure.

Cerebral oximetry data are presented in Table 2. We examined the oximetry data for the entire sample of 125 patients, the subset of 74 patients who had at least 1 episode of desaturation, and the subset of 56 patients who had at least 1 episode of desaturation and any postoperative cognitive outcome data. For the entire sample, baseline oximetry values did not differ significantly between groups. The

overall incidence of cerebral desaturation was 59%, with incidences of 66% in the intervention group and 53% in the control group and no significant difference. Duration of desaturation did not differ between the intervention and control groups. Median severity of desaturation was slightly greater in the intervention group because (1) a higher proportion (although not statistically significant) of the control group patients did not experience desaturation, and (2) the median is resistant to extreme values and obscures the fact that the control group had several cases of more severe desaturation (Q3 of severity are 66 vs 51 for the control group and intervention group, respectively). This difference did not remain statistically significant after adjusting for multiple comparisons.

Within the subgroup of patients who had at least 1 desaturation episode (Table 2), desaturation duration and severity were lower in the intervention group, but the differences were not statistically significant. Within the subgroup of patients with desaturation and postoperative cognitive testing, duration and severity were lower in the intervention group; desaturation duration did not differ significantly, but desaturation severity did. With log-transformation, the normality assumption was met, and *t* test results agreed with the nonparametric test results.

TABLE 3. Cognitive outcome comparisons of the intervention and control groups

Variable	Intervention		Control		ANCOVA	Difference Covariates-adjusted least- squares mean ^c (95% CI)	WGEE
	N	Mean (SD)	N	Mean (SD)	<i>P</i> value		<i>P</i> value
					Unadjusted ^a (FDR adjusted) ^b		Unadjusted (FDR adjusted) ^b
Change from T1 to T2							
Response speed [*]	37	0.21 (1.03)	54	−0.06 (0.89)	.739 (0.739)	0.03 (−0.01, 0.08)	.118 (0.236)
Processing speed [*]	38	0.19 (1.52)	52	−0.12 (0.70)	.415 (0.739)	0.05 (−0.03, 0.13)	.246 (0.328)
Attention	34	2.31 (1.72)	50	2.33 (1.68)	.611 (0.739)	−0.15 (−1.88, 1.58)	.865 (0.865)
Memory	38	0.44 (0.25)	54	0.08 (0.25)	.181 (0.724)	0.34 (−0.06, 0.74)	.097 (0.236)
Change from T1 to T3							
Response speed [*]	33	0.31 (0.92)	44	−0.04 (1.17)	.345 (0.460)	0.05 (0.01, 0.10)	.020 (0.040)
Processing speed [*]	33	−0.19 (0.62)	44	−0.15 (0.92)	.724 (0.724)	0.01 (−0.07, 0.08)	.901 (0.901)
Attention	31	1.95 (1.76)	41	0.66 (1.89)	.040 (0.080)	1.42 (−0.85, 3.68)	.220 (0.293)
Memory	34	0.60 (0.30)	44	−0.17 (0.33)	<.0001 (0.003)	0.65 (0.20, 1.10)	.005 (0.020)

Bold indicates $P < .05$. ANCOVA, Analysis of covariance; WGEE, weighted generalized estimating equation; SD, standard deviation; FDR, false discovery rate; CI, confidence interval. *Raw scores of response speed and processing speed were log-transformed before calculating change scores. T1 = preoperative testing, T2 = 3 months postoperative testing, T3 = 6 months postoperative testing. P values are based on a: analysis of covariance with baseline score, b: P value adjusted for false discovery rate, c: covariates-adjusted least squares means from the WGEE model with weight equals to the inverse probability of the subject dropping out at T2 and T3, respectively. The covariates are age, gender, education level, European System for Cardiac Operative Risk Evaluation, CPB time, and length of hospital stay. The same set of covariates was used for modeling the probability of dropping out for each individual at each visit (Appendix E1 and Tables E1–E4). At T2, 30 intervention group patients and 54 control group patients completed testing, whereas at T3, 44 intervention group patients and 34 control group patients completed testing; the N for each domain may be lower because of test scores that were flagged as invalid and removed from the data.

Neurocognitive Outcome Data

Table 3 presents the as-treated analysis comparing neurocognitive outcomes between the intervention and control groups. On the basis of the analysis of covariance, there were no significant differences between the intervention and control groups on any of the 4 cognitive factors at the 3-month postoperative testing. At the 6-month postoperative testing, the mean memory score was significantly better in the intervention group (0.60 [standard error, 0.30] vs

−0.17 [0.33], adjusted $P = .008$), and there was a trend for better mean attention score in the intervention group 6 months. On the basis of the WGEE mixed models, the log-transformed mean memory scores remained significantly higher in the intervention group at the 6-month postoperative time point (difference: 0.65 [0.23]; adjusted $P = .020$), and mean response speed became significant (0.05 [0.02], adjusted $P = .040$) (Appendix E1 shows model details).

TABLE 4. Clinical outcome comparisons of the intervention and control groups

Variable	Intervention		Control		P value
	N	Mean (SD) Median [Q1, Q3] N (%)	N	Mean (SD) Median [Q1, Q3] N (%)	
Death	59	2 (3.4)	66	1 (1.5)	.602 ^b
Hospital length of stay (d)	57	6 [5,7]	65	5 [5,7]	.999 ^a
ICU length of stay (d)	58	1 [1,2]	66	1 [1,2]	.999 ^a
Extubation (d)	58	1 [0, 1]	66	1 [0, 1]	.999 ^a
Time mechanical ventilation (h)	58	8.6 [6.8, 13.3]	66	7.9 [5.4, 10.9]	.311 ^a
ICU delirium	59	2 (3.4)	66	7 (10.6)	.170 ^b
SOFA on ICU admission	59	7 [5,9]	66	8 [5,9]	.885 ^a
ICU blood transfusion	59	10 (17.0)	66	9 (13.6)	.627 ^b
Bleeding requiring reoperation	59	2 (3.4)	66	4 (6.1)	.683 ^b
Renal failure	58	1 (1.7)	66	1 (1.5)	1.000 ^b
Respiratory failure	58	2 (3.4)	66	4 (6.1)	.684 ^b
Volume overload	58	44 (75.9)	66	50 (75.8)	1.000 ^b
Postoperative EF (%)	57	55 [46, 62]	66	55 [46, 60]	.926 ^a

P values are based on a: Kolmogorov–Smirnov test, b: Fisher exact test. SD, Standard deviation; ICU, intensive care unit; SOFA, Sequential Organ Failure Assessment; EF, ejection fraction.

TABLE 5. Summary of the literature examining the relationship between intraoperative near-infrared spectroscopy oximetry and cognitive outcomes in cardiac surgery patients

Article	Slater 2009	Colak 2014	de Tournay-Jetté 2011	Fudickar 2011
N	240	200	61	35
Surgery type	CABG	CABG	CABG ± CPB	CABG and/or valve
Other inclusion criteria	MMSE ≥24	Age 40-80 y	Age ≥65 y MMSE ≥24	MMSE ≥24
NIRS device	INVOS 5100BTM	INVOS 5100C	INVOS 4100	NIRO 300 Spectrometer
Study type	RCT	RCT	Prospective observational	Prospective observational
Design	Intervention at discretion of anesthesiologist vs blinded NIRS monitoring	Intervention by Denault protocol vs no NIRS monitoring	NIRS data revealed, Denault intervention protocol	NIRS monitoring with intervention
Intervention criteria	Decrease from baseline >20%	Decrease from baseline >20% or absolute value <50%	Decrease from baseline >20% or absolute value <50%	Decrease from baseline >20% or absolute value <55% for at least 1 min
Cognitive outcome measures	MMSE TMT-A TMT-B HVL GPB Stroop Test	MMSE CTT1 GPB	Rivermead Logical Memory* RAVLT* Digit Symbol TMT-A TMT-B Stroop Test Verbal fluency*	TMT Digit Symbol RAVLT Digit Span Verbal fluency
Preoperative test	Day of surgery	day of surgery	1 d before	1 d before
Postoperative test 1	Before hospital discharge	Day 7	Day 4-7	Day 5
Postoperative test 2	3 mo	NA	1 mo	NA
POCD definition	Decline ≥1 SD in at least 1 test score	MMSE score decrease ≥3 points or ≥1 SD increase in CTT 1 or GPB score	Decline ≥1 SD in at least 2 test scores	Decline >20% in at least 2 test scores
Overall incidence of POCD	60% before hospital discharge 29% at 3 mo	40.3%	81% at postoperative day 4-7 38% at 1 mo	43%
Result	Patients with prolonged desaturation had a significantly higher risk of early POCD	Incidence of early POCD was significantly lower in the INVOS interventional group (28%) than in the control group (52%) Prolonged desaturation was a risk factor for POCD	Patients with rSO ₂ <50% experienced more POCD at postoperative day 4-7 Decrease of >30% from baseline rSO ₂ was associated with POCD at 1 mo	SctO ₂ minimum during CPB correlated with POCD Absolute threshold of 65% during CPB discriminated patients with and without POCD

INVOS is manufactured by Medtronic, Minneapolis, Minn. NIRO is manufactured by Hamamatsu Photonics, Hamamatsu City, Japan. CABG, Coronary artery bypass grafting; CPB, cardiopulmonary bypass; MMSE, Mini-Mental State Exam; NA, not available; ASA, American Society of Anesthesiologists; RCT, randomized controlled trial; NIRS, near-infrared spectroscopy; TMT-A, Trail Making Test, Part A; TMT-B, Trail Making Test, Part B; HVL, Hopkins Verbal Learning Test; GPB, Grooved Pegboard test; CTT1, Color Trails Test 1; RAVLT, Rey Auditory Verbal Learning Test; BVRT, Benton Visual Retention Test; d2, d2-letter cancellation test; WAIS BD, Wechsler Adult Intelligence Scale Block Design subtest test; WAIS DS, Wechsler Adult Intelligence Scale Digit Span subtest; POCD, postoperative cognitive decline; SD, standard deviation. *Use of alternate forms.

For the secondary analysis, presence, duration, and severity of cerebral desaturation were not associated with cognitive change scores adjusting for covariates and group assignment (data not shown).

Perioperative Outcome Data

There were no differences between the intervention and control groups in perioperative outcomes (Table 4).

DISCUSSION

We conducted a prospective, randomized controlled trial in patients undergoing cardiac surgery with CPB to test the hypothesis that optimizing cerebral oxygenation using an intraoperative NIRS monitoring-guided treatment protocol results in better neurocognitive and perioperative outcomes. For the intervention group, intraoperative cerebral oxygen desaturation (rSO₂ <60% for >60 consecutive seconds at

TABLE 5. Continued

Hong 2008	Reents 2002	Yao 2004	Negargar 2007
100	47	101	72
Valve	CABG	CABG and/or valve	CABG ± CPB, valve
NA	NA	NA	ASA class II, III
INVOS 5100	INVOS 4100	INVOS 4100	INVOS 4100
Prospective observational	Prospective observational	Prospective observational	Prospective observational
Blinded NIRS monitoring, no intervention	Blinded NIRS monitoring, no intervention	Blinded NIRS monitoring, no intervention	NIRS monitoring with intervention
NA	NA	NA	Decrease from baseline >20%
MMSE TMT-A GPB	TMT-B BVRT d2 WAIS BD WAIS DS	MMSE	MMSE
1 d before	Day of surgery	Day of surgery	Day of surgery
Day 7	Day 6	Day 4-6	Day 1
NA	NA	NA	NA
decrease in MMSE score ≥ 3 points or $\geq 20\%$ increase in TMT-A or GPB score	decline ≥ 1 SD in at least 2 test scores	MMSE score decrease >20% or >1 SD	MMSE score decrease >20% and 1 SD
23%	34% (includes 6 patients with delirium)	18% (includes 2 patients with stroke)	11%
No difference in the incidence or duration of decrease in rSO ₂ between patients with vs without POCD	Duration and extent <2 cutoff values (40%, or decrease of >25% from baseline) was similar in patients with and without POCD	Area under threshold of <40% was a predictor of POCD	Higher incidence of prolonged desaturation in the POCD group vs no POCD group (33% vs 20%) Higher risk of POCD in patients with prolonged desaturation

either probe) triggered an intervention protocol; for the control group, the treatment team was unaware of the rSO₂ data and did not apply the treatment protocol. The intervention group had a significantly better outcome in memory scores at the 6-month postoperative testing period compared with the control group. There was no association between duration or severity of intraoperative cerebral desaturation and cognitive outcome. The groups did not differ with respect to perioperative outcomes.

These findings demonstrate that targeted therapy had a protective effect on neurocognitive outcome, but that presence, duration, and severity of cerebral desaturation per se were not associated with neurocognitive outcome. This implies that the treatment algorithm was an important aspect of the neuroprotection strategy, but further research is needed to indicate what specific aspects of the treatment algorithm, such as optimization of perfusion pressure, hemoglobin concentration, carbon dioxide

tension, or other factors, were contributory. Regarding the clinical significance of the observed differences in cognitive outcomes, changes of this magnitude are not expected to result in significant functional deficits, although they may signal mild insults to the brain that may be avoided.

Other studies have examined the relationship between intraoperative cerebral oximetry and neurocognitive outcomes in cardiac surgical patients.¹¹⁻¹⁸ Overall, the findings have been inconsistent, in part due to inter-study variations with respect to NIRS monitoring devices, definitions of desaturation, neurocognitive assessment methodologies, and definitions of POCD (Table 5). Likewise, a systematic literature review examining the relationship between intraoperative embolic load detected by transcranial Doppler ultrasound and cognitive outcome found that of 18 studies, 9 reported an association and 9 did not, and a systematic literature review examining the relationship between new ischemic brain lesions on diffusion-weighted magnetic resonance imaging and cognitive outcome found that of 18 studies reporting new ischemic lesions, 6 reported an association with cognitive outcome and 12 did not.^{19,20}

We chose the 60% threshold on the basis of the previous literature examining cerebral oximetry and outcomes after cardiac surgery, in which a range of thresholds have been applied but none higher than 60% to 65%.^{11,12,14,21-23} In addition, our own data in patients who have undergone aortic arch surgery showed statistically significant associations between desaturation duration and severe adverse outcome incidence for thresholds of 60% and 65%.²⁴ More specifically, every 10-minute increment of exposure time under the rSO₂ thresholds of 60% and 65% was associated with increased odds of severe adverse outcome by 34.6% (odds ratio, 1.346; 95% CI, 1.003~1.068; *P* = .038) and 17.4% (odds ratio, 1.6; 95% CI, 1.002~1.030; *P* = .025), respectively. For this study, we defined the criterion for intervention as rSO₂ less than 60% for more than 60 consecutive seconds at either probe so as not to have an overly aggressive protocol that would have made clinical interventions too burdensome for the clinical team.

With regard to the neurocognitive test data, we used continuous change scores rather than categorizing scores because such methods have been criticized on the grounds that they are arbitrary and without a theoretic basis, result in varying incidences of POCD within the same clinical sample depending on the applied criteria, and erroneously identify “POCD” in healthy nonsurgical study subjects based on score declines that are due to normal test–retest variability.²⁵

Study Limitations

There were several limitations to our study. To maximize the likelihood of triggering an effective NIRS-guided

treatment algorithm, we chose a mild threshold for cerebral desaturation compared with other investigations, with the goal of treating all potentially injurious cerebral desaturation episodes. As a consequence, the specificity of the defined threshold for cerebral desaturation was likely reduced, which could explain the lack of association of cerebral desaturation with outcome, because some episodes could have been clinically insignificant. As with all investigations, the NIRS data reflect saturation of cerebral tissue oxygenation from a small area of cortex within the frontal lobes, rather than brain structures that are most vulnerable to hypoxia (ie, hippocampi, watershed zones).

The internet-based cognitive testing protocol also was associated with limitations. Although this technology allowed for long-term postoperative testing in patients who live a long distance from the tertiary care medical center, it also required that patients access the test remotely from their home computers and take the time to complete the test. Early termination of the study decreased the statistical power and prevented us from exploring potential associations of the individual components of the treatment algorithm with neurocognitive outcome. Another study limitation was the differential follow-up with postoperative cognitive testing between the intervention and control groups.

Intraoperative NIRS monitoring is increasingly used routinely in the anesthetic, surgical, and perfusion management during cardiac surgery. It is also used during a variety of noncardiac surgical procedures. A systematic review found that noncardiac surgical procedures associated with pronounced decreases in rSO₂ are thoracic surgery involving single lung ventilation, major abdominal surgery, hip surgery, laparoscopic surgery with the patient in the reverse Trendelenburg position, shoulder arthroscopy in the beach chair position, and carotid endarterectomy with clamping of the internal carotid artery.²⁶ Investigations into the relationship between intraoperative cerebral oximetry and neurocognitive outcomes in noncardiac surgery have also yielded inconsistent findings.²⁷⁻³⁵

CONCLUSIONS

Additional prospective research is needed to define the role of cerebral oximetry–targeted therapy as a neuroprotection strategy for patients undergoing CPB and to create an evidence base to define guidelines for use of cerebral oximetry in cardiac surgical patients and other surgical patients at risk of cerebral injury.

Conflict of Interest Statement

Authors have nothing to disclose with regard to commercial support.

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Key Words: near-infrared spectroscopy, cardiac surgery, postoperative cognitive decline, neurocognitive outcomes, regional cerebral oxygen saturation

APPENDIX E1. PRIMARY DATA ANALYSIS

For the cognitive data, raw change scores from the preoperative and postoperative time points (ie, T1 to T2, and T1 to T3) were compared between the groups using the analysis of covariance method that controlled for the baseline score. The cognitive change scores were modeled as a linear function of group assignment and baseline score; $\text{change score} = \beta_0 \text{ (intercept)} + \beta_1 \text{ (Intervention group)} + \beta_2 \text{ (baseline score)}$. For the longitudinal data, the WGEE method was used. The models included the main effects of time (T1, T2, and T3) and group, and the time·group interaction, adjusting for the covariates of age, gender, education level, baseline euroSCORE, CPB time, and length of hospital stay (ie, $\text{cognitive score} = \beta_0 + \beta_1 [\text{intervention group}] + \beta_2 [\text{T2}] + \beta_3 [\text{T3}] + \beta_4 [\text{T2} \cdot \text{intervention}] + \beta_5 [\text{T3} \cdot \text{intervention}] + \text{covariates adjustment}$). A heterogeneous autoregressive structure was used for the working covariance matrix across the baseline and 3- and 6-month postoperative data; however, hypothesis testing was based on the empirical standard errors. The WGEE is often used for longitudinal data in which some subjects are lost to follow-up. Specifically, at each visit the probability for an individual to drop out at that time was estimated. The inverse of this probability was then incorporated as a numeric weight at the observation level when analyzing the data using the WGEE method. The same set of covariates as mentioned above was used to estimate the drop-out probability at each visit (ie, the missingness model). Intervention effects by visit were estimated from the model based on the least-squares means. The analyses were repeated in 2 ways: (1) excluding 6 patients with more than 10% of NIRS data that were missing and imputed, and (2) including the desaturation status and other covariates.

SECONDARY DATA ANALYSIS

A secondary analysis of the cognitive data was performed to evaluate the relationship between cognitive change scores and oximetry data. In this analysis, also using the WGEE method, the first step was to test the interaction between the treatment assignment and presence of cerebral desaturation. The purpose of this was to check whether the effect of intraoperative cerebral desaturation on cognitive change scores was modified by the treatment assignment, after adjusting for the covariates. If the interaction was not significant, the next step was to evaluate the independent effects of cerebral desaturation on the cognitive change scores. Last, the analysis was repeated using duration and severity of desaturation as predictors of cognitive change scores, independent of group assignment. *P* values were adjusted for multiple comparisons for the changes of 4 cognitive outcomes at each follow-up visit using the false discovery rate method.¹⁰

These adjustments do not necessarily control the family-wise error rate, but false discovery rate–controlling methods are more powerful and more liberal, and thus reject more null hypotheses than adjustments protecting the family-wise error rate.

SECONDARY DATA ANALYSIS RESULTS

For the secondary analysis, the WGEE models did not find evidence of synergistic effect on cognitive change scores between cerebral desaturation and treatment assignment, after adjusting for baseline cognitive score, age, education level, CPB time, and length of hospital stay (data not shown). Presence, duration, and severity of cerebral desaturation were not associated with cognitive change scores adjusting for covariates and group assignment. Excluding the 6 patients with more than 10% imputed NIRS data yielded almost identical quantitative results and led to the same qualitative conclusion.

TABLE E1. Weighted GEE model for log response speed

Parameter	Estimate	Standard error	95% Confidence limits	Z	Pr > Z
Parameter estimates for response model with empirical standard error estimates					
Intercept	−0.3514	0.1072	−0.5615 to −0.1414	−3.28	0.0010
Intervention	−0.0649	0.0287	−0.1211 to −0.0088	−2.27	0.0235
T2	−0.0013	0.0120	−0.0247 to 0.0222	−0.11	0.9148
T3	−0.0186	0.0161	−0.0501 to 0.0130	−1.15	0.2485
Intervention*T2	0.0340	0.0217	−0.0086 to 0.0766	1.56	0.1176
Intervention*T3	0.0541	0.0232	0.0086 to 0.0997	2.33	0.0199
Age	0.0020	0.0011	−0.0001 to 0.0042	1.83	0.0668
Female	0.0009	0.0247	−0.0475 to 0.0494	0.04	0.9695
Education (y)	−0.0064	0.0061	−0.0183 to 0.0055	−1.06	0.2901
euroSCORE	−0.0040	0.0057	−0.0151 to 0.0071	−0.70	0.4822
CPB (min)	−0.0001	0.0002	−0.0005 to 0.0004	−0.25	0.8055
Length of hospital stay (d)	0.0065	0.0042	−0.0017 to 0.0146	1.55	0.1209
Parameter estimates for missingness model					
Intercept	0.0255	2.2023	−4.2910 to 4.3420	0.01	0.9908
Age	0.0028	0.0263	−0.0488 to 0.0544	0.11	0.9154
Female	0.3965	0.6438	−0.8652 to 1.6583	0.62	0.5379
euroSCORE	0.1115	0.1659	−0.2136 to 0.4367	0.67	0.5014
Education (y)	0.1628	0.1034	−0.0399 to 0.3655	1.57	0.1155
CPB (min)	−0.0053	0.0051	−0.0153 to 0.0047	−1.04	0.2979
Length of hospital stay (d)	−0.0131	0.0759	−0.1618 to 0.1356	−0.17	0.8632

GEE, Generalized estimating equation; euroSCORE, European System for Cardiac Operative Risk Evaluation; CPB, cardiopulmonary bypass.

TABLE E2. Weighted GEE model for log processing speed

Parameter	Estimate	Standard error	95% Confidence limits	Z	Pr > Z
Parameter estimates for response model with empirical standard error estimates					
Intercept	0.9010	0.1223	0.6614 to 1.1407	7.37	<0.0001
Intervention	−0.0352	0.0393	−0.1122 to 0.0418	−0.90	0.3701
T2	−0.0357	0.0159	−0.0668 to −0.0045	−2.24	0.0249
T3	−0.0235	0.0325	−0.0873 to 0.0402	−0.72	0.4695
Intervention*T2	0.0486	0.0419	−0.0335 to 0.1308	1.16	0.2461
Intervention*T3	0.0051	0.0410	−0.0752 to 0.0855	0.12	0.9010
Age	0.0092	0.0021	0.0051 to 0.0132	4.43	<0.0001
Female	−0.0400	0.0392	−0.1168 to 0.0368	−1.02	0.3069
Education (y)	−0.0116	0.0072	−0.0257 to 0.0026	−1.60	0.1091
euroSCORE	−0.0048	0.0043	−0.0133 to 0.0037	−1.11	0.2683
CPB (min)	0.0003	0.0003	−0.0004 to 0.0010	0.87	0.3861
Length of hospital stay (d)	0.0037	0.0033	−0.0027 to 0.0101	1.14	0.2563
Parameter estimates for missingness model					
Intercept	0.4406	2.2230	−3.9164 to 4.7976	0.20	0.8429
Age	−0.0048	0.0280	−0.0596 to 0.0501	−0.17	0.8648
Female	0.3117	0.6527	−0.9675 to 1.5909	0.48	0.6329
euroSCORE	0.0914	0.1589	−0.2200 to 0.4027	0.58	0.5653
Education (y)	0.1804	0.1101	−0.0354 to 0.3962	1.64	0.1014
CPB (min)	−0.0059	0.0053	−0.0163 to 0.0046	−1.10	0.2720
Length of hospital stay (d)	−0.0253	0.0723	−0.1671 to 0.1165	−0.35	0.7267

GEE, Generalized estimating equation; euroSCORE, European System for Cardiac Operative Risk Evaluation; CPB, cardiopulmonary bypass.

TABLE E3. Weighted GEE model for attention

Parameter	Estimate	Standard error	95% Confidence limits	Z	Pr > Z
Parameter estimates for response model with empirical standard error estimates					
Intercept	4.4860	4.1362	−3.6207 to 12.5927	1.08	0.2781
Intervention	0.6807	0.8605	−1.0058 to 2.3671	0.79	0.4289
T2	2.2936	0.4832	1.3465 to 3.2406	4.75	<0.0001
T3	0.8573	0.7101	−0.5344 to 2.2490	1.21	0.2273
Intervention*T2	−0.1499	0.8843	−1.8831 to 1.5834	−0.17	0.8654
Intervention*T3	1.4186	1.1557	−0.8466 to 3.6838	1.23	0.2197
Age	0.0201	0.0420	−0.0621 to 0.1024	0.48	0.6317
Female	−1.3992	0.9084	−3.1796 to 0.3812	−1.54	0.1235
Education (y)	0.5694	0.1732	0.2300 to 0.9087	3.29	0.0010
euroSCORE	0.1979	0.1417	−0.0798 to 0.4756	1.40	0.1625
CPB (min)	0.0004	0.0069	−0.0131 to 0.0140	0.06	0.9504
Length of hospital stay (d)	−0.2122	0.0788	−0.3667 to −0.0577	−2.69	0.0071
Parameter estimates for missingness model					
Intercept	−1.1938	2.2759	−5.6545 to 3.2669	−0.52	0.5999
Age	0.0056	0.0272	−0.0478 to 0.0589	0.20	0.8378
Female	0.4162	0.6498	−0.8574 to 1.6898	0.64	0.5218
euroSCORE	0.1148	0.1619	−0.2024 to 0.4321	0.71	0.4780
Education (y)	0.1883	0.1061	−0.0196 to 0.3962	1.77	0.0759
CPB (min)	−0.0017	0.0057	−0.0129 to 0.0095	−0.29	0.7701
Length of hospital stay (d)	0.0003	0.0840	−0.1643 to 0.1649	0.00	0.9970

GEE, Generalized estimating equation; euroSCORE, European System for Cardiac Operative Risk Evaluation; CPB, cardiopulmonary bypass.

TABLE E4. Weighted GEE model for memory

Parameter	Estimate	Standard error	95% Confidence limits	Z	Pr > Z
Parameter estimates for response model with empirical standard error estimates					
Intercept	8.5135	0.6790	7.1826 to 9.8444	12.54	<0.0001
Intervention	−0.1192	0.2407	−0.5910 to 0.3526	−0.50	0.6204
T2	0.1527	0.0948	−0.0332 to 0.3385	1.61	0.1073
T3	−0.0461	0.1606	−0.3610 to 0.2687	−0.29	0.7740
Intervention*T2	0.3367	0.2026	−0.0604 to 0.7338	1.66	0.0965
Intervention*T3	0.6483	0.2300	0.1975 to 1.0990	2.82	0.0048
Age	−0.0112	0.0093	−0.0294 to 0.0070	−1.21	0.2266
Female	0.2198	0.1586	−0.0911 to 0.5306	1.39	0.1658
Education (y)	0.0350	0.0372	−0.0380 to 0.1080	0.94	0.3479
euroSCORE	−0.0168	0.0295	−0.0746 to 0.0411	−0.57	0.5698
CPB (min)	−0.0050	0.0018	−0.0084 to −0.0015	−2.80	0.0051
Length of hospital stay (d)	−0.0210	0.0322	−0.0842 to 0.0421	−0.65	0.5139
Parameter estimates for missingness model					
Intercept	0.6338	2.1491	−3.5782 to 4.8459	0.29	0.7680
Age	−0.0005	0.0256	−0.0506 to 0.0496	−0.02	0.9850
Female	0.4297	0.6382	−0.8212 to 1.6806	0.67	0.5008
euroSCORE	0.1382	0.1728	−0.2004 to 0.4768	0.80	0.4238
Education (y)	0.1324	0.1002	−0.0641 to 0.3288	1.32	0.1866
CPB (min)	−0.0056	0.0050	−0.0154 to 0.0041	−1.13	0.2587
Length of hospital stay (d)	−0.0143	0.0726	−0.1565 to 0.1279	−0.20	0.8438

GEE, Generalized estimating equation; euroSCORE, European System for Cardiac Operative Risk Evaluation; CPB, cardiopulmonary bypass.